GENERATIVE AI

GENERATIVE - AI

It is a part of the Data science in which content would be generated.

1. Text
2. Image
3. Video
4. audio

ChatGPT

- **ChatGPT** => Chat + GPT4
- **GPT** => Generative Pretrained Transformer
- It generates text based on the question what we ask.

Meta in WhatsApp

- It can generate text as well as images

In Machine Learning

- Models like RF, GBM, DT, and LR have sizes measured in kilobytes (KBs).
- Large Language Models (LLMs) generally have sizes in gigabytes (GBs), with basic models around 3GB.

LLMs deal with large amounts of language text data. We'll explore them in two parts:

1. Using this available LLMs
2. Making changes to this LLMs

Making use of LLMs are called **Prompt engineering**

- Prompt is the way we ask questions to the LLMs
- LLM models are very generic. we can ask any kind of questions.

It is so generic sometimes there is an ambiguity in answering questions.

This kind ambiguity can be deal by writing better prompt means by providing the correct context information in the prompt

Example - Provide me details of transformer
As an electrical engineer provide me details of the transformer in two sentences
As a Generative AI engineer provide me details of the transformer in two paragraphs

Prompt Structure

1. Providing the role by which the LLM to respond
2. Question
3. Response type what we are expecting

The role gives proper context to the LLM

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Types of Prompts

1. Basic prompts
2. Sentiment prompts
3. Translation prompts
4. Named entity recognition prompts
5. Summarization
6. Document QA
7. Chain of thought

1. Basic prompts

- **Example -** As movie analyst provide me the sentiment of the movie as positive, negative or neutral from the review in the text

Text:""" A cop trying to trace a gangster reaches a village where the sea is called Laal Samundar he is asked to meet Bahira. But Bhaira attacks him and villagers refuse to corporate until an old man Singappa comes to his aid the cop sees that underneath the sea there are lot many dead bodies which shocks him. Singappa tells him that the sea was ruled by Devara who along with Bahira and other neighbouring villagers would loot containers under the eye of customs for Murgan and Tulsi without knowing what it contains as it was livelihood for their villagers. But Devara finds that the looted containers would have arms and ammunition used to cause destruction and was also responsible for attacks in his village. Devara decides that further he will stop the loot and also not allow anyone to do but Bahira and others go against him. Devara stays as a guard at the sea coast not allowing anyone to enter. But one fine day Bahira and others plan to eliminate Devara and loot a ship but Devara defeats all of them but goes missing since then. Years later his family is waiting for his return while Bahira is waiting for revenge."""

2. Sentiment prompts

- **Example -** As movie analyst provide me the sentiment of the movie as positive, negative or neutral from the review in the text

Text:""" We don't need to look beyond Indian epics for riveting, complex stories of good versus evil, superheroes and demons, darkness and dawn. In Kalki 2898 AD, director Nag Ashwin attempts an unusual meeting point of the past and the future by blending stories from the Mahabharata with a futuristic, dystopian science fiction world. He underlines that heroes are not born, but rise. The masterstroke lies in casting the indefatigable Amitabh Bachchan as Ashwatthama, the oldest human alive from the Kurukshetra war, and making him the backbone of the narrative. He is pitted against Bhairava (Prabhas), to set the stage for the arrival of Kalki, the final avatar of Vishnu. The film takes giant leaps of faith and has awe-inducing segments. Is Kalki 2898 AD a game-changer? In terms of ambition, yes. In terms of writing and storytelling, there are niggles."""

3. Translation prompts

- **Example – 1**

As an English, Telugu Language specialist provide me translation of the below text into Telugu

Text:""" We don't need to look beyond Indian epics for riveting, complex stories of good versus evil, superheroes and demons, darkness and dawn. In Kalki 2898 AD, director Nag Ashwin attempts an

unusual meeting point of the past and the future by blending stories from the Mahabharata with a futuristic, dystopian science fiction world. He underlines that heroes are not born, but rise. The masterstroke lies in casting the indefatigable Amitabh Bachchan as Ashwatthama, the oldest human alive from the Kurukshetra war, and making him the backbone of the narrative. He is pitted against Bhairava (Prabhas), to set the stage for the arrival of Kalki, the final avatar of Vishnu. The film takes giant leaps of faith and has awe-inducing segments. Is Kalki 2898 AD a game-changer? In terms of ambition, yes. In terms of writing and storytelling, there are niggles."""

- **Example – 2**

As an English, Telugu Language specialist provide me translation of the below text into Telugu
Text:""" A cop trying to trace a gangster reaches a village where the sea is called Laal Samundar he is asked to meet Bahira. But Bhaira attacks him and villagers refuse to corporate until an old man Singappa comes to his aid the cop sees that underneath the sea there are lot many dead bodies which shocks him. Singappa tells him that the sea was ruled by Devara who along with Bahira and other neighbouring villagers would loot containers under the eye of customs for Murgan and Tulsi without knowing what it contains as it was livelihood for their villagers. But Devara finds that the looted containers would have arms and ammunition used to cause destruction and was also responsible for attacks in his village. Devara decides that further he will stop the loot and also not allow anyone to do but Bahira and others go against him. Devara stays as a guard at the sea coast not allowing anyone to enter. But one fine day Bahira and others plan to eliminate Devara and loot a ship but Devara defeats all of them but goes missing since then. Years later his family is waiting for his return while Bahira is waiting for revenge."""

4. Named entity recognition prompts

- **Example – 1** : As a movie analyst provide me named entities from the below text
Text:""" A cop trying to trace a gangster reaches a village where the sea is called Laal Samundar he is asked to meet Bahira. But Bhaira attacks him and villagers refuse to corporate until an old man Singappa comes to his aid the cop sees that underneath the sea there are lot many dead bodies which shocks him. Singappa tells him that the sea was ruled by Devara who along with Bahira and other neighbouring villagers would loot containers under the eye of customs for Murgan and Tulsi without knowing what it contains as it was livelihood for their villagers. But Devara finds that the looted containers would have arms and ammunition used to cause destruction and was also responsible for attacks in his village. Devara decides that further he will stop the loot and also not allow anyone to do but Bahira and others go against him. Devara stays as a guard at the sea coast not allowing anyone to enter. But one fine day Bahira and others plan to eliminate Devara and loot a ship but Devara defeats all of them but goes missing since then. Years later his family is waiting for his return while Bahira is waiting for revenge."""

- **Example – 2 :** As a movie analyst provide me named entities from the below text
Text:""" Today morning in Madhapur I visited Vijetha super market and purchased 2kgs of onions 1kg of tomatoes, 3 polo candies"""

- **Example – 3 :** As a movie analyst provide me named entities from the below text
Text:""" Today morning I am in Ramanasoft at Ameerpet and I am learning Generative AI"""

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5. Summarization

- **Example – 1 -** As movie analyst provide me summary in two sentences from the review in the text

Text:""" A cop trying to trace a gangster reaches a village where the sea is called Laal Samundar he is asked to meet Bahira. But Bhaira attacks him and villagers refuse to corporate until an old man Singappa comes to his aid the cop sees that underneath the sea there are lot many dead bodies which shocks him. Singappa tells him that the sea was ruled by Devara who along with Bahira and other neighbouring villagers would loot containers under the eye of customs for Murgan and Tulsi without knowing what it contains as it was livelihood for their villagers. But Devara finds that the looted containers would have arms and ammunition used to cause destruction and was also responsible for attacks in his village. Devara decides that further he will stop the loot and also not allow anyone to do but Bahira and others go against him. Devara stays as a guard at the sea coast not allowing anyone to enter. But one fine day Bahira and others plan to eliminate Devara and loot a ship but Devara defeats all of them but goes missing since then. Years later his family is waiting for his return while Bahira is waiting for revenge."""

6. Document QA

- **Example – 1 -** As movie analyst provide me list of actors who acted in the movie review in the text

Text:""" A cop trying to trace a gangster reaches a village where the sea is called Laal Samundar he is asked to meet Bahira. But Bhaira attacks him and villagers refuse to corporate until an old man Singappa comes to his aid the cop sees that underneath the sea there are lot many dead bodies which shocks him. Singappa tells him that the sea was ruled by Devara who along with Bahira and other neighbouring villagers would loot containers under the eye of customs for Murgan and Tulsi without knowing what it contains as it was livelihood for their villagers. But Devara finds that the looted containers would have arms and ammunition used to cause destruction and was also responsible for attacks in his village. Devara decides that further he will stop the loot and also not allow anyone to do but Bahira and others go against him. Devara stays as a guard at the sea coast not allowing anyone to enter. But one fine day Bahira and others plan to eliminate Devara and loot a ship but Devara defeats all of them but goes missing since then. Years later his family is waiting for his return while Bahira is waiting for revenge."""

7. Chain of thought

- **Example - 1**

On Monday I visited Ratna deep supermarket purchased 6 apples, on Monday Ramu ate one apple, on Tuesday Raghu took 2 apples, on Wednesday Hari purchased 12 apples, how many apples remaining now

- **Example - 2**

In January I have 20 leaves, for January I took 5 days leave, in May I took 6 days leave, in September I got a credit of 20 leaves, how many leave balance I have now.